

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and validated **R-Shiny programs** for statistical analysis and reporting of clinical trial data, processing approximately **15,000 patient records** per study.
- Collaborated with biostatisticians and cross-functional teams to interpret study requirements and ensure the accurate implementation of statistical analyses, contributing to **3 major clinical trial reports**.
- Prepared and reviewed statistical analysis plans (SAPs) and programming specifications, ensuring alignment with **ICH/GCP guidelines** for data integrity.
- Mentored **two junior programmers** on advanced R and R-Shiny techniques, improving their code quality and adherence to best practices.
- Executed statistical programming tasks using **Python** for data manipulation and analysis, integrating with existing clinical data management systems.
- Managed data integrity and compliance with industry regulations by implementing rigorous data validation checks and maintaining detailed audit trails throughout the programming lifecycle.
- Containerized **R-Shiny applications** using Docker for consistent deployment across development, testing, and production environments on AWS EC2 instances.

SKILLS

Programming Languages – R, R-Shiny, Python, SAS, SQL

Statistical Methodologies – Statistical Analysis, Clinical Trial Analysis, Data Interpretation, Regulatory Standards, ICH/GCP Guidelines

Data Management & Analysis – Clinical Data Management Systems, Data Integrity, Data Validation, Derived Datasets

Cloud & Containerization – AWS, Docker, EC2

Collaboration & Mentoring – Cross-functional Teams, Biostatisticians, Junior Programmers, Mentoring, Best Practices

PROJECTS

Clinical Trial Data Visualization Platform

- Developed an interactive **R-Shiny application** to visualize key performance indicators and statistical outputs for Phase III clinical trials.
- Implemented custom DAX measures for complex data aggregation and filtering, enabling stakeholders to explore trial results through **12 distinct dashboards**.
- Integrated **Power Query** for ETL processes, transforming raw clinical data into a structured format suitable for analysis and visualization.

Automated Statistical Reporting Suite

- Engineered a Python-based suite that automates the generation of standard statistical reports from clinical trial databases, reducing manual effort by **approximately 40%**.
- Wrote **SQL scripts** for data extraction and pre-processing from PostgreSQL databases, ensuring data accuracy for downstream analysis.
- Containerized the reporting applications using Docker, facilitating seamless deployment and execution on AWS cloud infrastructure.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)